

Lab 02 - First Verilog

In this lab, you've learned how to do an initial and simple design in Verilog to learn the Vivado tooling and process involved in RTL/FPGA design.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

The main purpose of this lab was to figure out how to create and run a simulated programmable program in Vivado to make a circuit work. We used this program to switch on a single light with a switch, `sw[0]`.

Lab Questions

1 - Describe the stages of building a Verilog project in Vivado.

First we begin by creating a verilog project. Run the simulation of the program files to verify they work and run correctly. Write the code needed for the board to work. Simulate the code with the hardware/circuit board to see how it would work in real time. Flip the switch so see if we can get the light to turn on or not.

2 - What is the value in looking at the elaborated design schematic?

It gives you a good idea of what you really need to to make a circuit board run. It shows all of the individual pieces in an easy to understand format.

3 - Why should we simulate our designs frequently? What does the simulation do?

We design our programs frequently to make sure it is updated and able to run correctly. The simulations help see whether our program will actually work with our hardware in real time or not.

Code Submission

<https://github.com/ashtonfarley/ECE-230L>